

- 3** HIV protease is an enzyme composed of two identical polypeptide chains. Each polypeptide chain is 99 amino acids long. During translation, the amino acids are joined by peptide bonds to form the polypeptide chain.

(a) Describe how a polypeptide of HIV protease is produced by the process of translation.

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(b) The synthesised polypeptide has a primary protein structure and can form a tertiary protein structure after translation. There are more bond types in the tertiary structure.

Compare the peptide bond formed during translation with the types of bond made during tertiary structure formation.

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- (c) Some polypeptides are translated as part of one long polypeptide chain. After translation, enzymes cut the polypeptide into separate functioning proteins.

HIV protease cuts a polypeptide that has been produced within the host cells of actively replicating HIV. The separate proteins are required in the replication of the virus.

Fig. 3.1 shows how the two polypeptide chains of HIV protease form an enzyme with an active site enclosed by flaps, forming the flap region.

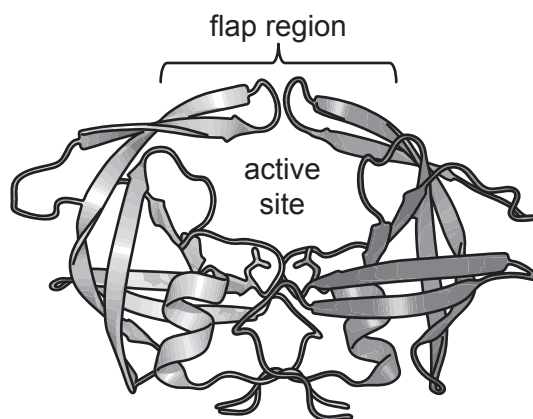


Fig. 3.1

- (i) The flap region of HIV protease is flexible.

With reference to Fig. 3.1, suggest **and** explain how the mechanism of action of HIV protease can be described as an induced fit.

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- (ii) Indinavir is one of the therapeutic drugs used in HIV anti-retroviral therapy (ART). It is similar to the polypeptide substrate of HIV protease.

Suggest **and** explain how indinavir acts as a therapeutic drug.

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